Treehouse Design

rivt Report

Example report

Proj. 0001

Contents

01.1 Summary

This report covers the structural design of a treehouse in Novato, California. The design follows the requirements of the California Building Code (CBC).

The treehouse is supported by a single Douglas fir beam spanning 12 feet between two trees. The beam supports a uniformly distributed load (UDL) from the treehouse floor and live load from occupants. The design includes calculations for the required beam size and material properties to ensure safety and compliance with building codes.

01.2 Load Combinations and Geometry

Dead and live load contributions to beam UDL.

Table 1: ASCE 7-05 Load Effects

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

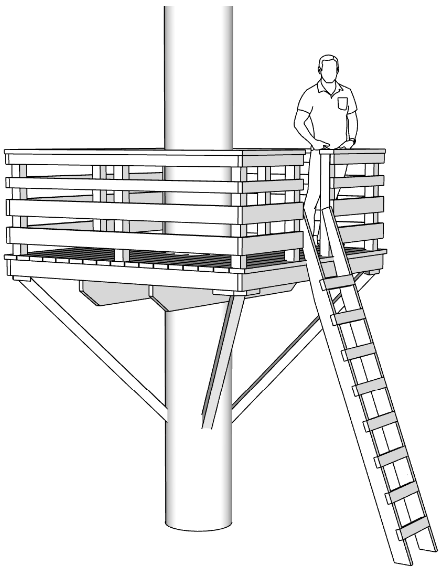

Fig. 1 - Treehouse

02.1 Loads

Successive value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 1: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL

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D_4	3.00 klf	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-O5 LL
b_1	10.00 inch	254.00 mm	beam width
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
Fb_1	20000.00 lb_in2	137.90 MPA	allowable stress

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02.1 Table test

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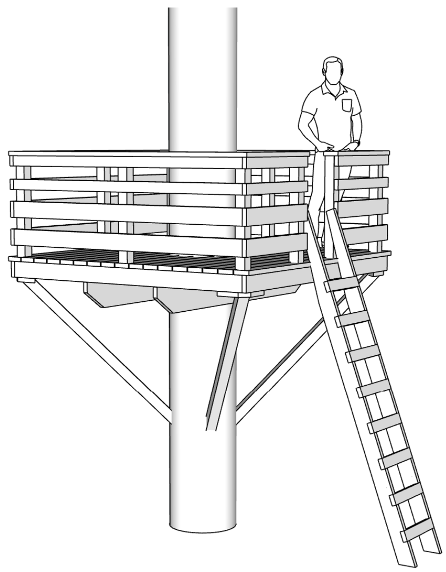

Fig. 1 - Treehouse